Experiment 14

To capture, save, and analyze network traffic on TCP / UDP / IP / HTTP / ARP /DHCP /ICMP /DNS using Wireshark Tool

Aim

To understand Wireshark packet capture and encapsulation at the layers of a protocol stack.

Requirements

Wireshark/TShark, capture support, active lab interface and traffic generators.

Procedure / algorithm

1. Choose the active Ethernet interface and configure capture stopping after 100 packets. Save the captured packets.

2. Inspect Packet List, Packet Details and Packet Bytes. Expand link, IP and transport/application headers.

3. Apply the supplied TCP/UDP, ARP, DNS, HTTP, IP/ICMP and DHCP display filters. Record the addresses and fields actually present.

4. Generate lab DNS/web/ping/DHCP traffic as needed. For TCP/UDP and DNS, inspect Statistics > Flow Graph; for TCP, use Follow > TCP Stream.

5. Distinguish capture filters, which select stored traffic, from display filters, which select visible saved traffic.

6. For this submission, capture only project-generated localhost TCP/UDP echo and localhost ICMP. Analyse the resulting 100-packet file with TShark.

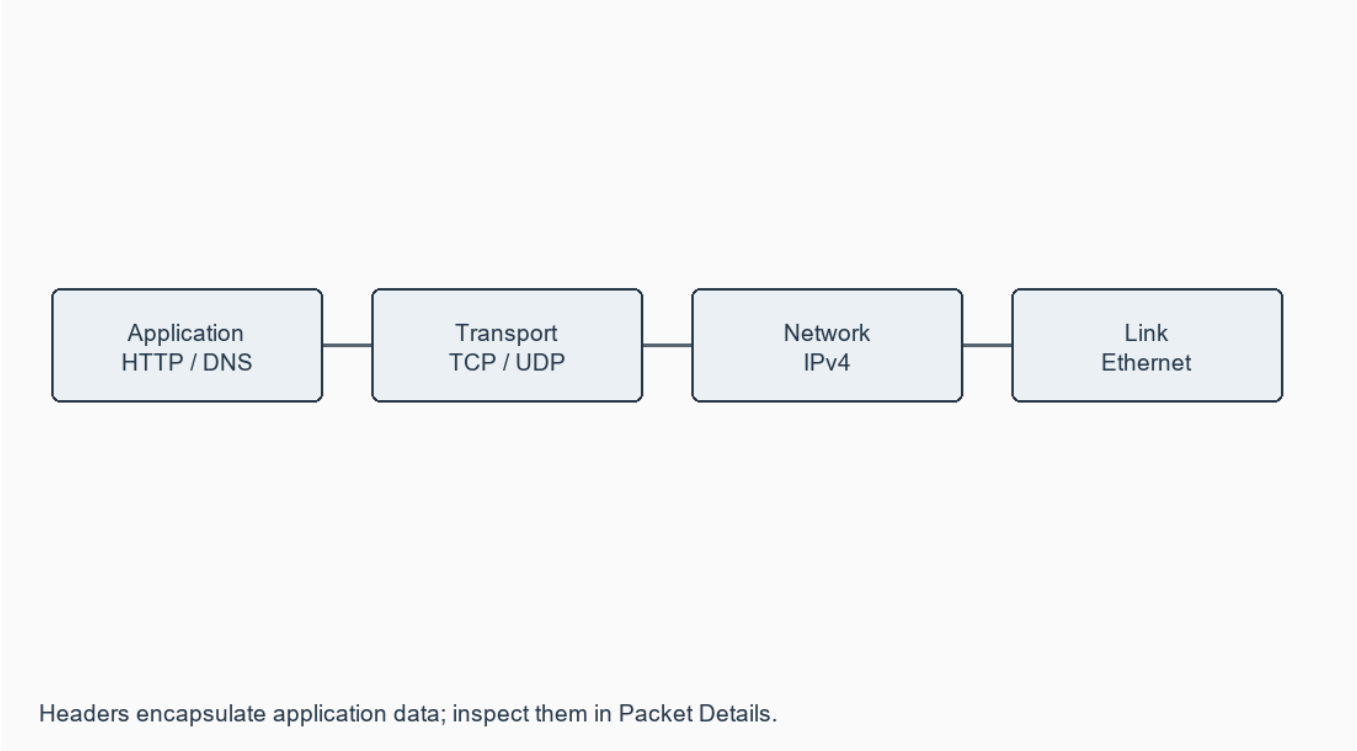
Complete source / configuration: capture_commands.txt

```
GUI: Capture > Options > active Ethernet interface > stop after 100 packets.
Generate assigned lab traffic, save capture.pcapng, then use display_filters.txt.
Statistics > Flow Graph; Follow > TCP Stream.
CLI for assigned Ethernet interface:
tshark -D
tshark -i INTERFACE_NUMBER -c 100 -w capture.pcapng
tshark -r capture.pcapng -Y "tcp or udp"
tshark -r capture.pcapng -Y arp
tshark -r capture.pcapng -Y dns
tshark -r capture.pcapng -Y http
tshark -r capture.pcapng -Y "ip or icmp"
tshark -r capture.pcapng -Y "udp.port == 67 or udp.port == 68"
tshark -r capture.pcapng -q -z io,phs
```

Complete source / configuration: display_filters.txt

```
tcp or udp
arp
dns
http
ip or icmp
udp.port == 67 or udp.port == 68
ip.addr == 127.0.0.1
tcp.port == 50990
udp.port == 50991
```

Diagram: encapsulation



Actual output / verification

```
ACTUAL TSHARK ANALYSIS OF 100 LIVE LOOPBACK PACKETS
Own TCP/UDP echo traffic and ICMP to 127.0.0.1 only.
This is loopback, not the manual's physical Ethernet capture.
ARP, DNS, HTTP and DHCP lab captures remain pending.

Frame   Source  Destination Protocol
1   127.0.0.1   127.0.0.1   ICMP
2   127.0.0.1   127.0.0.1   ICMP
3   127.0.0.1   127.0.0.1   ICMP
4   127.0.0.1   127.0.0.1   ICMP
5   127.0.0.1   127.0.0.1   UDP
6   127.0.0.1   127.0.0.1   UDP
7   127.0.0.1   127.0.0.1   UDP
8   127.0.0.1   127.0.0.1   UDP
9   127.0.0.1   127.0.0.1   UDP
10  127.0.0.1   127.0.0.1   UDP
11  127.0.0.1   127.0.0.1   UDP
12  127.0.0.1   127.0.0.1   UDP
13  127.0.0.1   127.0.0.1   UDP
14  127.0.0.1   127.0.0.1   UDP
15  127.0.0.1   127.0.0.1   UDP
16  127.0.0.1   127.0.0.1   UDP
17  127.0.0.1   127.0.0.1   UDP
18  127.0.0.1   127.0.0.1   UDP
19  127.0.0.1   127.0.0.1   UDP
20  127.0.0.1   127.0.0.1   UDP
21  127.0.0.1   127.0.0.1   UDP
22  127.0.0.1   127.0.0.1   UDP
23  127.0.0.1   127.0.0.1   UDP
24  127.0.0.1   127.0.0.1   UDP
25  127.0.0.1   127.0.0.1   UDP
26  127.0.0.1   127.0.0.1   UDP
27  127.0.0.1   127.0.0.1   TCP
28  127.0.0.1   127.0.0.1   TCP
29  127.0.0.1   127.0.0.1   TCP
30  127.0.0.1   127.0.0.1   TCP
31  127.0.0.1   127.0.0.1   TCP
32  127.0.0.1   127.0.0.1   TCP
33  127.0.0.1   127.0.0.1   TCP
34  127.0.0.1   127.0.0.1   TCP
35  127.0.0.1   127.0.0.1   TCP
36  127.0.0.1   127.0.0.1   TCP
37  127.0.0.1   127.0.0.1   TCP
38  127.0.0.1   127.0.0.1   TCP
39  127.0.0.1   127.0.0.1   TCP
40  127.0.0.1   127.0.0.1   TCP
41  127.0.0.1   127.0.0.1   TCP
42  127.0.0.1   127.0.0.1   TCP
43  127.0.0.1   127.0.0.1   TCP
44  127.0.0.1   127.0.0.1   TCP
45  127.0.0.1   127.0.0.1   TCP
46  127.0.0.1   127.0.0.1   TCP
47  127.0.0.1   127.0.0.1   TCP
48  127.0.0.1   127.0.0.1   TCP
49  127.0.0.1   127.0.0.1   TCP
50  127.0.0.1   127.0.0.1   TCP
51  127.0.0.1   127.0.0.1   TCP
52  127.0.0.1   127.0.0.1   TCP
53  127.0.0.1   127.0.0.1   TCP
```

```
54  127.0.0.1   127.0.0.1   TCP
55  127.0.0.1   127.0.0.1   TCP
56  127.0.0.1   127.0.0.1   TCP
57  127.0.0.1   127.0.0.1   TCP
58  127.0.0.1   127.0.0.1   TCP
59  127.0.0.1   127.0.0.1   TCP
60  127.0.0.1   127.0.0.1   TCP
61  127.0.0.1   127.0.0.1   TCP
62  127.0.0.1   127.0.0.1   TCP
63  127.0.0.1   127.0.0.1   TCP
64  127.0.0.1   127.0.0.1   TCP
65  127.0.0.1   127.0.0.1   TCP
66  127.0.0.1   127.0.0.1   TCP
67  127.0.0.1   127.0.0.1   TCP
68  127.0.0.1   127.0.0.1   TCP
69  127.0.0.1   127.0.0.1   TCP
70  127.0.0.1   127.0.0.1   TCP
71  127.0.0.1   127.0.0.1   TCP
72  127.0.0.1   127.0.0.1   TCP
73  127.0.0.1   127.0.0.1   TCP
74  127.0.0.1   127.0.0.1   TCP
75  127.0.0.1   127.0.0.1   TCP
76  127.0.0.1   127.0.0.1   TCP
77  127.0.0.1   127.0.0.1   TCP
78  127.0.0.1   127.0.0.1   TCP
79  127.0.0.1   127.0.0.1   TCP
80  127.0.0.1   127.0.0.1   TCP
81  127.0.0.1   127.0.0.1   TCP
82  127.0.0.1   127.0.0.1   TCP
83  127.0.0.1   127.0.0.1   TCP
84  127.0.0.1   127.0.0.1   TCP
85  127.0.0.1   127.0.0.1   TCP
86  127.0.0.1   127.0.0.1   TCP
87  127.0.0.1   127.0.0.1   TCP
88  127.0.0.1   127.0.0.1   TCP
89  127.0.0.1   127.0.0.1   TCP
90  127.0.0.1   127.0.0.1   TCP
91  127.0.0.1   127.0.0.1   TCP
92  127.0.0.1   127.0.0.1   TCP
93  127.0.0.1   127.0.0.1   TCP
94  127.0.0.1   127.0.0.1   TCP
95  127.0.0.1   127.0.0.1   TCP
96  127.0.0.1   127.0.0.1   TCP
97  127.0.0.1   127.0.0.1   TCP
98  127.0.0.1   127.0.0.1   TCP
99  127.0.0.1   127.0.0.1   TCP
100 127.0.0.1   127.0.0.1   TCP
```

Actual TShark protocol summary

```
===================================================================
Protocol Hierarchy Statistics
Filter:

frame                                    frames:100 bytes:5022
  null                                   frames:100 bytes:5022
    ip                                   frames:100 bytes:5022
      icmp                               frames:4 bytes:256
      udp                                frames:22 bytes:930
        data                             frames:22 bytes:930
      tcp                                frames:74 bytes:3836
        data                             frames:36 bytes:2140
===================================================================
```

Expected output - not an execution record

TCP shows SYN/SYN-ACK/ACK and data; ARP shows address resolution; DNS query/response; ICMP Echo Request/Reply; DHCP Discover/Offer/Request/ACK. HTTPS does not expose ordinary unencrypted HTTP fields.

Observations

Actual evidence is a 100-packet Npcap loopback capture, not physical IEEE 802.3 Ethernet. It contains ICMP, UDP and TCP. ARP/DNS/HTTP/DHCP captures and GUI screenshots are pending. ARP has no transport header; DNS commonly uses UDP 53 (also TCP); HTTP uses TCP 80. Promiscuous mode sees only traffic delivered to the interface.

Result

A real 100-packet local capture was saved and analysed; remaining manual Ethernet/protocol exercises are documented but not claimed as executed.